

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-7674
Project Name	AI FAQ ASSISTANT API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to text-based FAQs only	Cannot answer complex or out-of-syllabus queries	High
2	Supports only 3 languages (Eng, Hindi, Tamil)	Many regional users cannot use it	Medium
3	Small knowledge base (PDFs only)	Answer accuracy drops for new queries	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles 50-100 users at a time	Use Cloud deployment (AWS) + Load Balancer to handle 10,000+ users
2	Data Storage	Data stored in local files / SQLite	Move to Vector Database (Pinecone) + MongoDB for large FAQ data
3	Performance	Response time 2-3 sec	Use faster LLM API + Caching + Optimized NLP model to get <1 sec response
4	Security	Basic security only	Add User Authentication, Data Encryption and Role-based Admin Access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Voice Assistant in 5+ Indian Languages	3 Months	More users can use it easily
Phase 2	Integration with College ERP and WhatsApp Business API	6 Months	Fully automated admission support
Phase 3	AI Self-Learning from user feedback + Analytics Dashboard	9 Months	Bot will improve automatically without manual update